CSCI1409 Deep Learning | Aidyn Abirov | Week 7 Tasks 1–3 – Estimated time: about 2 hours – **Deadline:** before the next lecture – **Submit:** notebook via MS Teams Assignments

This homework is about **recurrent neural networks**. You will run an RNN by hand and from scratch, see why its memory fades, open up an LSTM cell to see how its gates keep memory alive, and finally train an RNN and an LSTM that invent new first names, one letter at a time. Everything happens in the provided notebook `CSCI1409_Week07_Homework.ipynb`: fill in every `# TODO`, and use the `assert` lines to check your work.

Rules

- Work individually.
- **Tasks 1–2: no AI** (search engines, lecture notes, textbooks, and the PyTorch documentation are fine – list references). Do not use AI-generated search summaries.
- **Task 3: AI allowed** only to explain concepts or help debug Task 3 code. State which tool you used and for what. You must understand and be able to defend every submitted line.
- Do not change the `assert` lines. Your from-scratch functions must not call `nn.RNN`, `nn.LSTM`, or `nn.LSTMCell`.
- Use PyTorch, NumPy, and Matplotlib only.
- Submit `Surname_Firstname_Homework07.ipynb` via the Assignments tab on Teams; restart the kernel and run all cells first. No submissions once the next lecture starts.

Setup. Same as Homeworks 5–6 (`pip install torch matplotlib`, or Google Colab). Task 3 downloads a list of 32,000 names (about 230 KB) the first time. A GPU is not needed: all training takes well under a minute on a laptop CPU.

Task 1 – An RNN step by step

About 30 minutes. No AI.

1. **By hand:** run the tiny RNN $h_t = \tanh(x_t + 0.5\, h_{t-1})$ on the sequences $(1, 0, 0)$ and $(0, 0, 1)$ – same numbers, different order.
2. **From scratch:** write `rnn_forward` with one loop over time; it must match your hand results and PyTorch's `nn.RNN`.
3. **Shapes and parameters:** fill in the shapes of `output` and `h_n`, and count the parameters of an RNN and an LSTM by hand.
4. Answer Questions 1.

Task 2 – Why do we need LSTMs?

About 30 minutes. No AI.

1. **Fading memory by hand:** compute w^{T-1} for three values of w and two lengths, and see where it vanishes or explodes.
2. **The LSTM cell by hand:** compute the new cell state and hidden state for four gate settings (keep, overwrite, hide, mix), then code `lstm_update`.
3. **A full LSTM step from scratch:** compute the four gates from the weights and match PyTorch's `nn.LSTMCell`.
4. **Measure it:** plot how much the first input still affects the last hidden state, for an RNN, a default LSTM, and an LSTM with an open forget gate.
5. Answer Questions 2.

Task 3 – A name generator

About 45 minutes. AI assistance allowed, with individual defense.

A character-level language model reads one letter at a time and predicts the next one.

1. **Data:** join all names into one text, cut it into chunks of 20 characters, and build targets shifted by one character (check one chunk by hand).
2. **Model:** complete `forward` – one-hot letters → `nn.RNN` or `nn.LSTM` → `Linear(128, 27)` at every step.
3. **Training:** complete the sequence loss and the training loop; train an RNN and an LSTM for 5 epochs each.
4. **Generation:** complete `generate` – sample a letter, feed it back in, and repeat until the end symbol.
5. **Look inside:** next-letter probabilities after different beginnings, and the hidden state while reading a name.
6. Answer Questions 3. *Optional:* sampling temperature.

In the defense, be ready to explain any line of your code, for example: why `output[:, -1]` equals `h_n`, which slice of the bias is the forget gate, and why the target is the input shifted by one.

Submission checklist

- Every `# TODO` filled, every `assert` passes, and the notebook runs from a fresh kernel (Restart & Run All)
- All **Your answer** cells written (Questions 1, 2, 3)
- Hand calculations for 1a, 1c, 2a, 2b, and 3a done on paper first
- References listed; Task 3 AI use recorded (or “no AI used”); no AI used for Tasks 1–2
- Submitted via Teams Assignments before the next lecture begins

Reminder: a working notebook does not replace the individual defense during practice.